Slot Game Programming Test

Objective

Create a basic 3x3 slot game based on the provided reference.

The goal of this assessment is to evaluate:

- Programming fundamentals
- Code quality and maintainability
- Architecture and design decisions
- Project organization (folder structure, prefab usage, naming conventions, and scene hierarchy organization)
- Problem-solving skills
- Implementation of core slot game mechanics, including payline evaluation and win detection
- Creativity and technical initiative in designing and implementing gameplay mechanics

---

Technical Requirements

Platform

- Engine: Cocos Creator 3.8.4
- Language: TypeScript
- Resolution: 720 × 1280 (Portrait)

Slot Configuration

- 3 Reels (Columns)
- 3 Visible Rows
- Each reel must contain at least 8 symbols

Core Features

Spin Button

- Starts a spin sequence for all reels
- Generates a result and stops the reels accordingly

Turbo Mode

Implement 3 speed levels:

Level	Behavior
1	Normal speed
2	Fast speed
3	Instant result

Win Detection

Support the following paylines:

- Top row
- Middle row
- Bottom row
- Top-left → Center → Bottom-right
- Top-right → Center → Bottom-left

Win Presentation

- Visually indicate winning paylines
- Play a simple animation to indicate matched symbols

Code Quality & Architecture

- Follow SOLID principles or a hybrid architecture approach where appropriate
- Use clean and maintainable code
- Organize systems with clear separation of responsibilities
- Avoid unnecessary coupling between components
- Design the project to be easy to understand, maintain, and extend

Project Organization

- Use prefabs appropriately to promote reusability and maintainability
- Organize the scene hierarchy in a clear and logical structure
- Maintain consistent naming conventions for nodes, prefabs, scripts, and assets
- Structure project folders in a way that is easy to navigate and extend

---

Optional Features

Applicants are encouraged to showcase their creativity, technical expertise, and understanding of game systems by implementing additional gameplay features beyond the required scope.

Examples include:

- Reel skip functionality
- Additional paylines
- Auto-spin
- Win multipliers
- Wild symbols
- Scatter symbols
- Bonus rounds
- Free spins
- Cascading wins
- Symbol upgrades
- Progressive rewards
- Combo systems
- Variable reel configurations
- Additional win patterns
- Custom game rules

These are only examples. Applicants are free to implement their own ideas and mechanics.

Deliverables

Provide the following:

Source Code Repository

- GitHub, GitLab, or Bitbucket repository
- Include meaningful commit history

Setup Instructions

- Steps required to run the project

Technical Notes

Briefly explain:

- Architecture approach used
- Key design decisions
- Assumptions made during development

- Any optional features or improvements implemented

Evaluation Criteria

Evaluation Area	Description
Functionality	Correct implementation of the required features
Code Quality	Readability, maintainability, and consistency
Architecture	Appropriate use of SOLID principles or hybrid architecture patterns
Project Organization	Folder structure, prefab usage, naming conventions, and scene hierarchy organization
Problem Solving	Quality of technical decisions and implementation approach
Creativity & Initiative	Additional gameplay mechanics, system design, and overall craftsmanship
Version Control	Meaningful commit history and repository management

---

Scope

Focus on functionality rather than visual polish.

The following are not required:

- Production-ready visuals
- Advanced animations
- Complex UI design
- Commercial-grade game features

Applicants may add additional functionality if they wish.

Timeline

Submission deadline: 3 days upon receipt of this document.

Questions

For clarifications, contact:

michael.dominise@ntt.com.ph

Good luck!